

$$\textcircled{1} \text{ a) } \mu_n \frac{\partial \psi_n(x)}{\partial x} + \sigma_+ \psi_n(x) = \sum_{k=0}^K \frac{2k+1}{2} \sigma_{s,k} P_k(\mu_n) \sum_{m=1}^N w_m P_m(\mu_n) \psi_m(x) \\ + \frac{\mu_{n+1/2} - \mu_{n-1/2}}{2} Q(x)$$

b) $\textcircled{1}$ take l^{th} moment of $\mathcal{F} S_N$: $\int P_l(\cdot)$

$$\textcircled{2} \int \mu P_l \frac{\partial \psi(\mu, x)}{\partial x} + \int \sigma_+ P_l \psi(x, \mu) = \sum_{k=0}^K \frac{2k+1}{2} \sigma_{s,k} \phi_k \int P_l P_k \\ \rightarrow + \int P_l \left[\frac{\mu_{l+1/2} - \mu_{l-1/2}}{2} Q(x) \right]$$

$$\textcircled{3} \frac{2}{2x} \left[\int \frac{l+1}{2l+1} P_{l+1} \psi(\mu, x) + \frac{l}{2l+1} P_{l-1} \psi(\mu, x) \right] + \sigma_+ \phi_l = \\ \rightarrow \sum_{k=0}^{K+1} \frac{2k+1}{2} \sigma_{s,k} \phi_k \delta_{k,l} \frac{2}{2l+1} + \frac{Q(x)}{2}$$

$$\textcircled{3} \frac{2}{2x} \left[\frac{l+1}{2l+1} \phi_{l+1}(x) + \frac{l}{2l+1} \phi_{l-1}(x) \right] + \sigma_+ \phi_l(x) = \sigma_{s,l} \phi_l(x) + \frac{Q(x)}{2}$$

(2) a) $p(\mu_c) = \frac{1}{2}$

$$\sigma_{S,1} = \int_{-1}^1 P_1(\mu) \sigma_S(\mu) d\mu = \sigma_S \int_{-1}^1 P_1(\mu) p(\mu) d\mu$$

$$p(\mu) d\mu = p(\mu_c) d\mu_c \rightarrow p(\mu) \frac{d\mu}{d\mu_c} = p(\mu_c) \frac{d\mu_c}{d\mu}$$

$$\mu = \frac{1 + A\mu_c}{\sqrt{A^2 + 1 + 2A\mu_c}} = \frac{1 + \mu_c}{\sqrt{2 + 2\mu_c}} = \frac{1}{\sqrt{2}} \sqrt{1 + \mu_c}$$

$$\mu_c = 2\mu^2 - 1$$

$$p(\mu) = p(\mu_c) \frac{d\mu_c}{d\mu} \rightarrow \frac{1}{2} \cdot \frac{d}{d\mu} (2\mu^2 - 1)$$

$$p(\mu) = 2\mu$$

$$0^{th} \rightarrow \sigma_{S,0} = \sigma_S \int_{-1}^1 \frac{1}{2} \cdot 2\mu d\mu = \left. \frac{\sigma_S \mu^2}{2} \right|_{-1}^1 = 0$$

$$\sigma_{S,1} = \sigma_S \int_{-1}^1 \mu \cdot 2\mu d\mu = \left. \sigma_S \cdot \frac{2}{3} \mu^3 \right|_{-1}^1 = \sigma_S \left[\frac{2}{3} + \frac{2}{3} \right] = \frac{4\sigma_S}{3}$$

no higher order terms, 2μ is perfectly expanded by 1st order polynomial

(b) c

RMSRE :

order	1 MeV	30 MeV
1	0.073 729	0.1748
3	0.029 121	0.70295
5	0.009 7831	0.03818
10	0.000 19995	0.55405
100	X	0.0000949

the smoother 1 MeV converged much quicker, while the near singular 30 MeV had oscillatory convergence and took 100 orders to converge



